

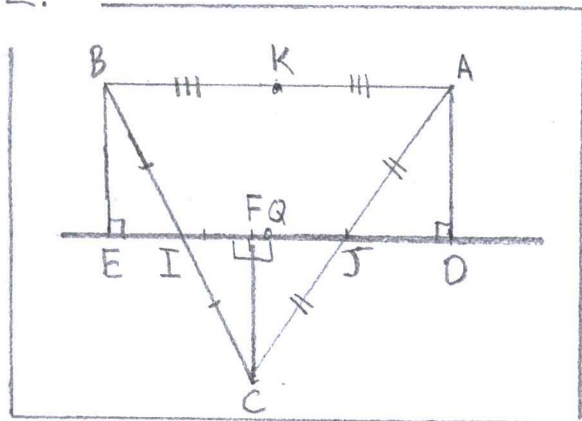
Chapter 6: The Discovery of Non-Euclidean Geometry

Exercises:

1. (from exercise) "The most important exercise"

- Every triangle has a circumscribed circle
- Wallis' postulate on similar triangles
- A rectangle exists
- Clavius' axiom that equidistant locus on one side of a line is the set of points on a line.
- Some triangle has angle sum equal to 180°
- An angle inscribed in a semi-circle is a right angle
- The Pythagorean law holds for right triangles.
- A line cannot lie entirely in the interior of an angle.
- Any point in the interior of an angle lies on a segment with endpoints on the sides of the angle
- Areas of triangles are unbounded

2. a) (from exercise)



"In any Hilbert plane, $\overline{AD} \cong \overline{CF} \cong \overline{BE}$,
hence $\square EDAB$ is a Saccheri
quadrilateral with base $\overline{ED}$,
Summit $\overline{AB}$ "

"Saccheri triangles
have equal content"

Proof: Hypothesis	Evidence
1) Triangle $\triangle ABC$ is ready for geometry	(pg 98) "Definition- Triangle"
2) I, J, K are midpoints of $\overline{BC}$, $\overline{CA}$, $\overline{AB}$, respectively	Proposition 4.3 (midpoints)
3) D, E, F are the feet of the perpendiculars from A, B, C to $\overline{IJ}$.	(pg 269) "Definition- medial line"
4) Foot of perpendicular from K to $\overline{IJ}$ was necessary as Q	(pg 269) "Definition- medial line"
5) $\square EDAB$ is a Saccheri Quadrilateral	RAA result by Converse

$$6) \triangle BKQ \cong \triangle AKQ$$

$$7) \angle B \cong \angle A$$

$$8) \triangle BKE \cong \triangle AKD$$

$$9) \angle EQK \cong \angle DQK$$

$$10) \angle EBQ \cong \angle DAQ$$

$$11) \triangle BQK \cong \triangle AQK$$

$$12) \angle BKQ \cong \angle AKQ$$

$$13) \angle BIE \cong \angle FIC \\ \cong \angle AJD$$

$$14) \triangle BEI \cong \triangle IFC$$

$$15) \overline{BE} \cong \overline{CF}$$

$$17) \overline{BE} \cong \overline{AD}$$

$$18) \overline{AD} \cong \overline{CF} \cong \overline{BE}$$

Proposition 3.22 (SSS)

Proposition 3.19 (Angles Add)

Axiom C-4 (SAS)

(Pg 19) "Definition-
right angle"

Proposition 3.23

Proposition 3.22 (SSS)

Euclid Postulate IV

Euclid Postulate I. XV

Proposition 4.1 (SAA)

Axiom C-6

(Pg 177) "Definition-
Saccheri"

Axiom C-2, RAA hypothesis

b) (from exercise) "Perpendicular bisector of $\overline{AB}$ is
also perpendicular to $\overline{IJ}$ "

Proof: Hypothesis

Evidence

1) Triangle $\triangle ABC$ is ready for geometry

(pg 98) "Definition-Triangle"

2) I, J, K are midpoints of $\overline{BC}, \overline{CA}$, and $\overline{AB}$

Proposition 4.3 (midpoints)

3) D, E, F can be feet of the perpendiculars from A, B, C to $\overline{IJ}$

(pg 269) "Definition-medial line"

4) Foot of the perpendicular from K to $\overline{IJ}$ is Q

(pg 269) "Definition-medial line"

6) $\overline{KQ} \perp \overline{IJ}$

Proposition 6.4

c) (from exercise) " $\overline{ED} = 2\overline{IJ}$ "

Proof: Hypothesis

Evidence

1) Triangle $\triangle ABC$ is ready for geometry

(pg 98) "Definition-Triangle"

2) I, J, K are midpoints of $\overline{BC}, \overline{CA}$, and $\overline{AB}$

Proposition 4.3 (midpoints)

3) D, E, F can be feet from the perpendiculars A, B, C to $\overline{IJ}$

(pg 269) "Definition-medial line"

4) From K to $\overline{IJ}$ is perpendicular Q .

5) $\square EDAB$ is Saccheri

$$6) \triangle BKQ \cong \triangle AKQ$$

$$7) \angle B \cong \angle A$$

$$8) \triangle BKE \cong \triangle AKD$$

$$9) \angle EQK \cong \angle DQK$$

$$10) \angle EBQ \cong \angle DAQ$$

$$11) \triangle BQK \cong \triangle AQK$$

$$12) \angle BKQ \cong \angle AKQ$$

$$13) \angle BIE \cong \angle FIC \\ \cong \angle AJD$$

$$14) \triangle BEI \cong \triangle IFC$$

$$16) \overline{EI} = \overline{IF}$$

$$17) \triangle FJC = \triangle JDA$$

$$18) \overline{ED} = \overline{EI} + \overline{IF} \\ + \overline{IJ} + \overline{FD} \\ = 2\overline{IF} + 2\overline{IJ} \\ = 2\overline{IJ}$$

(pg 269) "Definition - Medial line"

RAA result by converse

Proposition 3.22 (S.S.S.)

Proposition 3.19 (Angle Addition)

Axiom C-4 (S.A.S.)

(pg 19) "Definition - supplementary"

Proposition 3.23

Proposition 3.22 (S.S.S.)

Euclid Postulate IV

Euclid Postulate I. XV

Axiom C-6 (S.A.S.)

(pg 14) "Definition - congruent"

Axiom C-6 (S.A.S.)

Axiom C-3

RAA hypothesis

d) (from exercise) " K, F, C are collinear if and only if $\overline{AC} \cong \overline{BC}$ "

Proof Hypothesis	Evidence
1) Triangle $\triangle ABC$ is ready for geometry	(pg 98) "Definition - Triangle"
2) I, J, K are mid-points of $\overline{AB}, \overline{BC}$, and $\overline{CA}$	Proposition 4.3 (midpoints)
3) D, E , and F are feet to the perpendiculars from A, B, C to $\overline{IJ}$	(pg 269) "Definition - medial line"
4) Foot of the perpendicular from K to $\overline{IJ}$ is Q	(pg 269) "Definition - medial line"
5) $\overline{AC} \cong \overline{BC}$	RAA hypothesis
6) $Q = F$	RAA result
7) $K * F * C$	(pg 70) "Definition - collinear"

(From exercise) "F is the midpoint of CK iff the plane is Euclidean"

Proof :	Hypothesis	Evidence
1) With steps from prior stage...		RAA hypothesis
2) I and J are midpoints, so $\overline{IJ}$ is a midline		(pg 269) "Definition-medial line"
3) CK meets midline $\overline{IJ}$ at F, so F is a midpoint		RAA result

(from exercise) " K, F , and C are not collinear
and the plane is not Euclidean,
so $\overline{CF}$ is not perpendicular
to $\overline{AB}$ "

Proof: Hypothesis	Evidence
1) Triangle $\triangle ABC$ is ready for geometry	(pg 98) "Definition-Triangle"
2) I, J, K are mid-points of $\overline{BC}, \overline{CA}$, and $\overline{BA}$.	Proposition 4.3 (midpoints)
3) D, B, C are feet to the perpendiculars from A, B, C to $\overline{IJ}$	(pg 269) "Definition-medial line"
4) Foot of the perpendicular from K to $\overline{IJ}$ is Q	(pg 269) "Definition-medial line"
5) $\angle A + \angle B + \angle C < 180^\circ$	Basic Theorem 6.1
6) $\overline{CF}$ is not a right angle to AB .	Hilberts Hyperbolic Axiom of Parallels

e) (From exercise) "Pythagorean equation holds for all right triangles and if KC is a right angle then $\overline{AB} = 2\overline{IJ}$ "

Proof:	Hypothesis	Evidence
1) Triangle $\triangle ABC$ is ready for geometry.		(pg 98) "Definition - Triangle"
2) I, J, K are midpoints of $\overline{BC}, \overline{CA}, \overline{AB}$.		Proposition 4.3 (midpoints)
3) D, E, F are the feet of the perpendiculars from A, B, C to $\overline{IJ}$		(pg 269) "Definition - medial line"
4) Foot of perpendicular from K to $\overline{IJ}$ is Q		(pg 269) "Definition - medial line"
5) $\square EDAB$ is Saccheri		RAA result by converse
6) $\triangle BKQ \cong \triangle AKQ$		Proposition 3.22 (S.S.S.)
7) $\angle B \cong \angle A$		Proposition 3.19 (Angles Add)
8) $\triangle BKE \cong \triangle AKD$		Axiom C-4 (S.A.S.)
9) $\angle EBQ \cong \angle DAQ$		Proposition 3.23
10) $\triangle BQK \cong \triangle AQK$		Proposition 3.22 (SSS)
11) $\angle BKQ \cong \angle AKQ$		(pg 119) "Definition - congruent"
12) $\angle BIE \cong \angle FIC$ $\cong \angle AJD$		Euclid Postulate I. XV

$$13) \triangle BEI \cong \triangle IFC$$

$$14) \overline{BE} \cong \overline{CF}$$

$$15) \overline{BE} \cong \overline{AD}$$

$$16) \overline{AD} \cong \overline{CF} \cong \overline{BE}$$

$$17) \overline{AB} \cong \overline{ED}$$

$$18) \overline{AB} \cong 2\overline{IJ}$$

Proposition 4.1 (SAA)

(pg 14) "Definition - Congruent"

(pg 177) "Definition - Saccheri Quadrilateral"

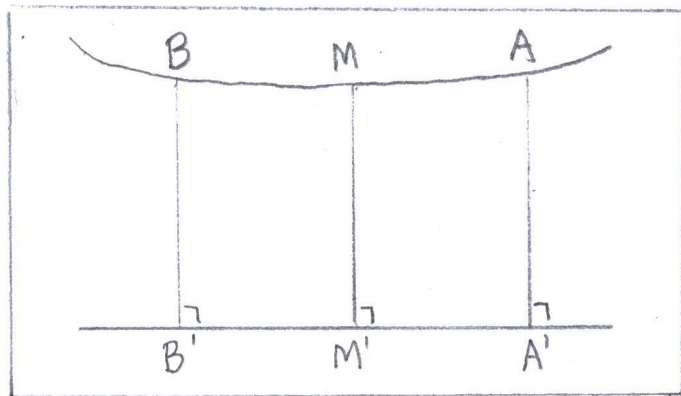
Axiom C-2

(pg 177) "Definition - Saccheri Quadrilateral"

RAA hypothesis

3.

(from exercise) Parallel lines l and l' have



"Common perpendicular"

a common perpendicular segment MM' ... and MM' is the shortest segment between any pair of l and any point of l' "

Proof: Hypothesis

Evidence

1) Rectangles do not exist in the plane

RAA hypothesis

2) $\overleftrightarrow{l} \parallel \overleftrightarrow{l'}$

Euclid's Postulate V

3) Point A on $\overleftrightarrow{l}$

(pg 269) "Definition - medial line"

has a foot A' on l'

4) Point B on $\vec{l}$ has
a perpendicular foot

B' on l'

5) $\overline{AA'} \cong \overline{BB'}$

6) $\square ABB'A$ is a
Saccheri Quadrilateral

7) $\angle AA'B = \angle B'BA$

8) $\vec{l}$ and $\vec{l}'$ have a
common perpendicular
segment $\overline{MM'}$ from
the midpoint M on $\overline{AB}$

9) $\overline{MM'}$ is the short-
est segment be-
tween $\vec{l}$ and $\vec{l}'$. Also
segments $\overline{AA'}$ and $\overline{BB'}$
increase as A, B
from M .

(pg 269) "Definition-
medial line"

RAA hypothesis #2

(pg) "Definition-
Saccheri"

Proposition 4.12

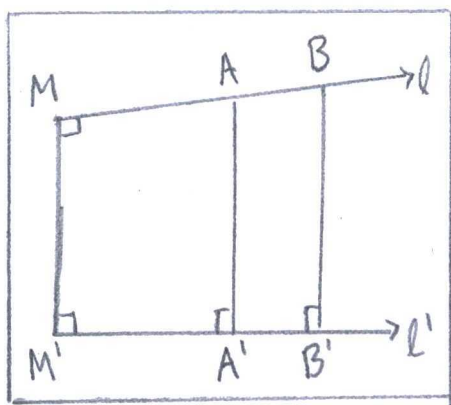
Alternate Interior
Angle Theorem 4.1

Corollary 1

RAA result.

4 (from exercise)

" $\overline{AA'} < \overline{BB'}$ "



"Common Perpendicular"

Hypothesis

1) Rectangles do not exist in the plane

2) $\vec{l}$ is a ray along $M \times A \times B$

3) $\vec{l}'$ is a ray along $M' \times A' \times B'$

4) $\overline{MM'}$ is a common perpendicular segment between $\vec{l}$ and $\vec{l}'$

5) $\square MMA'A'$ is a Lambert Quadrilateral where $\angle MAA' < 90^\circ$

6) $\angle AA'B$ is obtuse because supplement to $\angle MAA'$

7) $AA' < BB'$ because $\angle M < \angle AA'B$

Evidence

RAA hypothesis

(pg 18) "Definition-Rays"

(pg 18) "Definition-Rays"

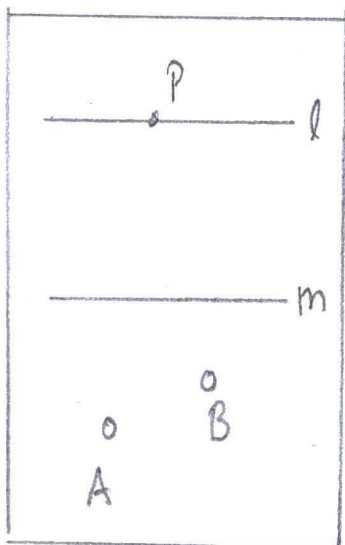
Proposition 6.4.

(pg 180) "Definition-Lambert Quadrilateral"

(pg 19) "Definition-Supplementary"

Proposition 4.13
RAA result

5

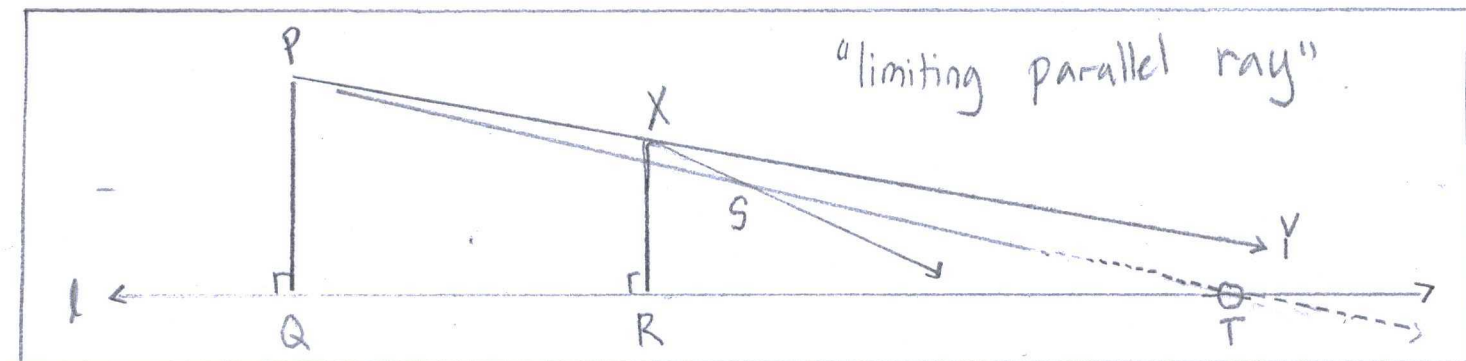


"Same Side"

(from exercise) "A and B are on the same side of l "

Proof: Hypothesis	Evidence
1) Lines $\vec{l}$ and $\vec{m}$ are parallel	Euclid's Postulate IV
2) Points A and B lie opposite sides of $\vec{m}$ to $\vec{l}$	Axiom B-4(ii)
3) Point P $\vec{l}$	(pg 12) "Definition- Incident"
4) P is opposite $\vec{m}$ to A and B	Axiom B-4(ii)
5) A and B are on the same side of $\vec{l}$	Axiom B-4(i)

6.



(from exercise) " $\overrightarrow{XY}$ is a limiting parallel ray to $\overleftrightarrow{l}$ through X "

Proof: Hypothesis

Evidence

1) Ray $\overrightarrow{XS}$ between $\overrightarrow{XY}$ and $\overrightarrow{XR}$ meets $\overleftrightarrow{l}$ at the perpendicular R from X to $\overleftrightarrow{l}$.

(pg 115) "Definition between"

2) S and Y are on the same side of $\overrightarrow{XR}$

Axiom B-4(ii)

3) P and Y are on opposite side of $\overrightarrow{XR}$

Axiom B-4(i)

4) S and Y are on the same side of $\overrightarrow{PQ}$

Axiom B-4(ii)

5) S and R are on

the same side of $\overline{PY}$
 6) Q and R are on
 the same side of $\overline{PY}$

7) Q and S are on the
 same side of $\overline{PY}$

8) Thus, $\overline{PS}$ lies between
 $\overline{PY}$ and $\overline{PQ}$, at
 point T on $\overleftrightarrow{l}$.

9) Point X is exterior
 to $\triangle PQT$

10) $\overline{XS}$ does not
 intersect PQ

11) Hence $\overline{XS}$ intersects
 $\overline{QT}$, so $\overline{XS}$ meets
 $\overleftrightarrow{l}$

Axiom B-4 (ii)

Axiom B-4 (ii)

Axiom B-4 (ii)

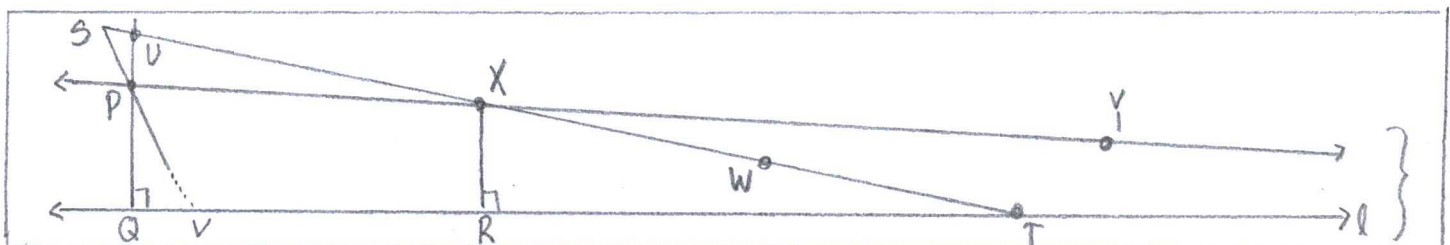
Axiom B-4 (ii)

(pg 117) "Definition-
 exterior"

(pg 18) "Definition-
 emanate from a
 vertex"

Proposition 3.9(a)

7.



"Limiting Parallel"

(From exercise) "Let us assume XY is a limiting parallel to $\vec{l}$, with $P * X * Y$. $\vec{PY}$ is a limiting parallel to $\vec{l}$ ".

Proof: Hypothesis	Evidence
1) By the data from questioning...	RAA hypothesis
2) Unique point S is in common to $\overline{PZ}$ and $\overline{XW}$, so $S * P * Z$	Proposition 2.1
3) $\overline{SX}$ meets $\overline{PQ}$ in a point U such $U * P * Q$	Proposition 2.1
4) Point W forms $U * X * W$	(pg 70) "Definition - collinear"
5) W and Y are on the same side of $\overline{XR}$	Axiom B-4(i)
6) P and Y are on	Axiom B-4(ii)

Opposite sides of $\overleftrightarrow{XR}$

7) W and Y are on
the same side $\overleftrightarrow{PQ}$

8) W and R are on
the same side

$$\overleftrightarrow{XY} = \overleftrightarrow{PY}$$

9) Q and R are on
the same side of
 $\overleftrightarrow{PY}$

10) Q and W are on
the same side of
 $\overleftrightarrow{PY}$

11) Thus, $\overleftrightarrow{XW}$ lies
between $\overleftrightarrow{XY}$ and
 $\overleftrightarrow{XR}$, so $\overleftrightarrow{XW}$ inter-
sects $\overleftrightarrow{QR}$ at point
T.

12) Point X is exterior
to $\triangle UQT$

Axiom B-4(i)

Axiom B-4(i)

*Key step because
PW less than limiting
parallel PY

Axiom B-4(i)

Axiom B-4(i)

(pg 115) "Definition -
between"

(pg 117) "Definition -
exterior"

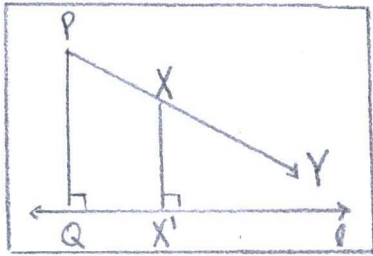
13) $\overline{XW}$ does not intersect $\overline{PQ}$

(pg) "Definition-emanating from vertex"

14) Hence $\overline{XW}$ intersects QT , then $\overline{XW}$ meets l

Proposition 3.9(a)

Q.



(from exercise) " $\overline{PQ} > \overline{XX'}$ "

"right limiting parallel"

Proof: Hypothesis

Evidence

1) $\overline{PX}$ be a right limiting parallel ray to l through P .

RAA hypothesis

2) Q and X' be the feet of the perpendiculars from P to l

Proposition 4.3

3) $\square PQXX'$

Proposition 4.12(b)

4) $\overline{QX} < \overline{PX}$

Corollary 3 to Proposition 4.13

5) $\overline{PX} < \overline{PX'}$

Corollary 3 to Proposition 4.13 because Hypotenuse $>$ triangle leg

$$\angle P = \angle XPX' + \angle X'PQ < 90^\circ$$

$$\text{So, } \angle X'PQ < 90^\circ$$

$$\angle QXX' = 90^\circ \text{ because supplement angle}$$

$$\text{8) Side } \overline{PX'} \text{ is largest on } \triangle PX'X, \text{ thus } \angle PXX' > 90^\circ$$

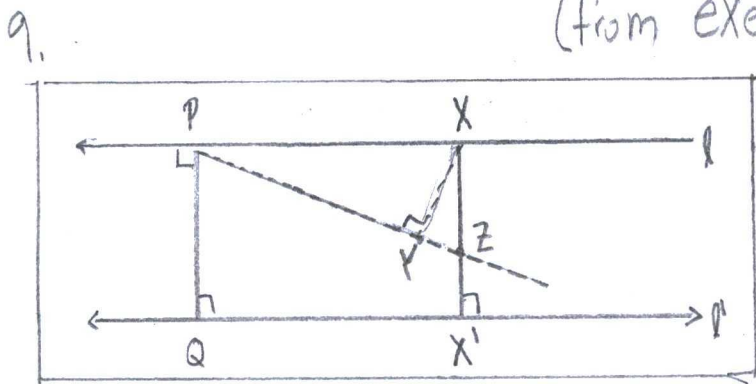
$$\text{9) } \overline{PQ} > \overline{XX'} \text{ because acute } \angle P \text{ and obtuse } \angle PXX'$$

(pg 129) "Definition-acute"

Measurement Theorem 4.3(A.5)

Proposition 4.5

Proposition 4.13



(from exercise) "As X recedes endlessly from P on l, the segment XX' increases indefinitely."

"Segment... increases indefinitely"

Proof: Hypothesis

1) Rectangles do not exist

Evidence

RAA hypothesis

$$2) l \parallel l'$$

3) $\overleftrightarrow{PQ}$ is a common perpendicular to l and l'

4) For a point X on l , X' is the foot of the perpendicular from l to l'

5) A point Z is on $\overline{XX'}$

6) $\overline{PZ}$ limits parallels

7) Y is a perpendicular foot between $\overline{PZ}$ and $\overline{XY}$

8) $\overline{XZ} > \overline{XY}$ because $\angle XYZ > \angle XZY$

9) $\overline{XX'} > XZ$

Euclid's Postulate V

Proposition 2.1

Proposition 4.1

Axiom B-2

(pg) "Definition-limiting parallels"

Proposition 4.1

Proposition 4.5

Aristotle's Angle...
Unbounded Axiom

10) If $\overline{PX}$ increased,
then $\overline{XX'}$ endlessly.

because $\angle XYZ = 90^\circ$

is always greater

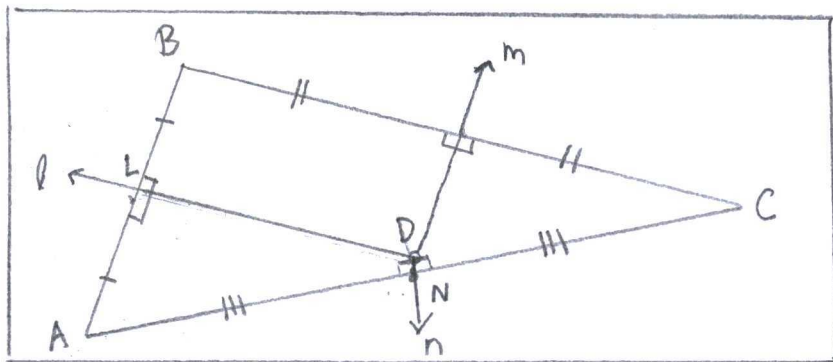
than $\angle XZY$, when

$$\overline{XX'} > \overline{XZ} > \overline{XY}$$

RAA result

10.

(from exercise) "Two perpendicular

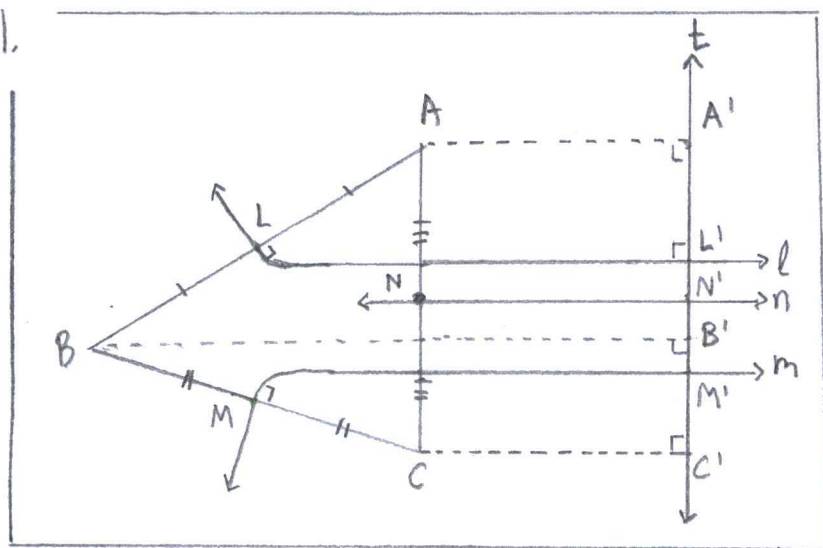
bisectors $\vec{l}$ and $\vec{m}$ have a common point."

"Perpendicular
bisectors... meet
at a point"

General Proof:

A triangle ($\triangle ABC$) has a midpoint (Proposition 4.3) with a perpendicular bisector (Proposition 4.4). The bisectors $\vec{l}$ and $\vec{m}$ each traverse toward a second side on the triangle (Pasch's Theorem), and cross at a common point.

11.



(from exercise)

"The third perpendicular
bisector $\vec{n}$ is also
a perpendicular to
 $\vec{t}$."

"Perpendicular bisectors of $\triangle ABC$ have
Common Perpendicular"

Proof: Hypothesis

Evidence

1) Lines $\vec{l}$ and $\vec{m}$
have a common
perpendicular bi-
sector $\vec{t}$.

Proposition 4.3

2) Line $\vec{n}$ is also
parallel to $\vec{l}$

Euclid's Postulate V

3) $\vec{n} \parallel \vec{l}$ since $\vec{n} \parallel \vec{l}$

Proposition 4.10

4) Points A and B
are on opposite
sides $\vec{l}$

Axiom B-4(i)

5) Points B and C
are on opposite
sides $\vec{m}$.

Axiom B-4(i)

6) $\triangle ABC$

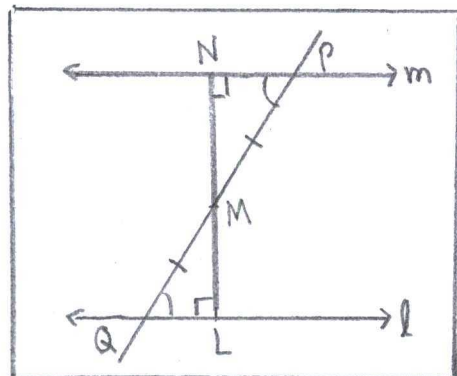
(pg 98) "Definition-
Triangle"

7) Points A' , B' , and
 C' are feet on $\vec{t}$
from perpendiculars
A.B and C.

Proposition 4.3.

12.

(pg 275) Theorem 4.1

"Divergently
Parallel"

"If alternate interior angles are
congruent, then the lines are parallel"

Proof: Hypothesis

Evidence

1) $\vec{l} \parallel \vec{m}$

Euclid's Postulate V

2) $\overline{PQ}$ crosses $\vec{l}$
and $\vec{m}$

Proposition 2.1

3) M is the mid-
point of $\overline{PQ}$ Proposition 4.3
(midpoints)4) N is the per-
pendicular foot
from M to m
and L is the
perpendicular foot
from M to l.(pg 269) "Definition-
medial line"

5) $\overline{MN} = \overline{ML}$

Proposition 4.3 (midpoints)

6) $\overline{MQ} = \overline{MP}$

Proposition 4.3 (midpoints)

7) $\angle QLM = \angle MNP$
 $= 90^\circ$

Euclid Postulate IV

$$8) \angle MQL = \angle MPN$$

$$9) L * M * N$$

$$10) \angle LQM = \angle NPM$$

Proposition 4.1

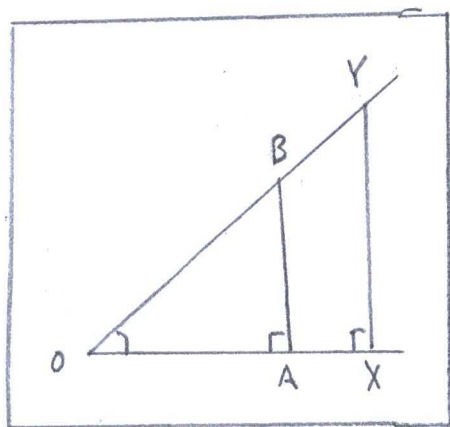
(pg 70) "Definition-collinear"

RAA result

and Proof through
Converse

(result $\rightarrow$ hypothesis)

13.



"Aristotle's
Axiom"

(from exercise) "Aristotle's axiom holds
in a hyperbolic plane."

Proof: Hypothesis

Evidence.

1) An acute angle
 $\angle XOY$

(pg 129) "Definition-acute"

2) $\overline{AB} \perp \overline{OX}$ and
 $\overline{XY} \perp \overline{OX}$

(pg 42) "Definition-perpendicular"

3) X is the perpendicular foot
to Y

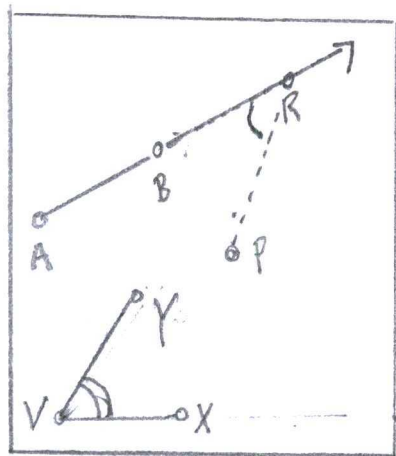
Proposition 4. —

4) $\square ABXY$ is a
bi-right quadrilateral,
 $\angle ABY > \angle XYB$
 $\Leftrightarrow \overline{XY} > \overline{AB}$

Proposition 4.13

14.

(from exercise) "A non-Euclidean Hilbert plane satisfies the important corollary to Aristotle's axiom."



" $\angle PRA < \angle XVY$ "

Proof: Hypothesis	Evidence
1) $\angle PRA < \angle XVY$	Measurement Theorem 4.3(B)
2) $\angle XVY + \angle VYX + \angle YXV \leq 180^\circ$	(pg 250) Acute angle hypothesis
3) $\angle PRA + \angle RAP + \angle APR \leq 180^\circ$	(pg 250) Acute angle hypothesis

15.

(from exercise) Saint Augustine

"The good christian should beware of mathematicians and all those who make empty prophesies. The danger already exists that all mathematicians have made a covenant with the devil to darken the spirit and to confine man in the bonds of Hell."

Personal comment, "Women in math cannot
be confined to the bonds
of Hell."